The question

MarinFold predicts contacts from a single sequence. The regime where that has something to prove is the one where MSA-based methods run out of signal: proteins with few homologs.

PR #257 produced our best decontaminated checkpoint (#232 `m2-p06` training, step 363,000) and scored it on `eval-val`, `eval-denovo`, and the legacy 554. This experiment finishes the read — `eval-test`, previously unscored for this checkpoint — and then bins every natural protein in the eval universe by the depth of the ColabFold MSA an MSA-based competitor would have had.

What ran

887 evaluation units on twelve single-H100 CoreWeave shards: the legacy 554 plus all 333 scorable FoldBench monomers. 100 rollouts per unit under the fixed #82 recipe, 88,700 usable, zero unfinished, 9m41s.

Depth comes off the two Modal volumes of ColabFold a3m files that Protenix's `+MSA` arm already ran with, measured through #74's `msa_depth.py` — raw sequence count and Neff, one code path for both volumes.

Validation

PR #257's published legacy-554, `eval-val`, and `eval-denovo` aggregates are the gate: same weights, same worker, so a disagreement would mean the path moved, not the model. All twelve reproduced, largest difference 0.0044.

The depth measurements match #247's independent count on all 314 FoldBench natural proteins exactly. Of the 11 stems both Modal volumes hold, 9 agree exactly and all 11 land in the same tier, so the three-week gap between the two ColabFold runs does not move the stratification.

The eval sets

subset	n	R (all)	R (long)
eval-test (first read for this checkpoint)	217	**0.5693**	0.5464
eval-val	97	0.5561	0.5402
eval-denovo	19	0.6110	0.5745
legacy 554	554	0.6059	0.5566

+0.032 on eval-test over the #232 sweep checkpoint, and 0.144 clear of the seq-KNN null over its own decontaminated corpus. Protenix-v2 + MSA scores 0.8446 on the same 217 proteins.

MSA depth — accuracy does not hold up

Mean all-range R-precision, 372 natural proteins:

depth	n	MarinFold	Protenix-v2 + MSA	ESMFold2	Protenix-v2 single-seq
<10	29	0.379	0.510	0.556	0.457
10–100	33	0.281	0.758	0.480	0.291
100–1000	77	0.416	0.817	0.671	0.283
≥1000	233	0.616	0.858	0.827	0.249

MarinFold degrades with MSA depth much as the MSA methods do — depth is a proxy for how well a family is represented anywhere, including in the AFDB corpus it trained on. Paired against Protenix-v2 single-seq, MarinFold's whole advantage is a deep-MSA phenomenon: +0.367 at ≥1000, and indistinguishable from zero in both shallow bins.

What survives

Two results, both paired per-protein with bootstrap intervals:

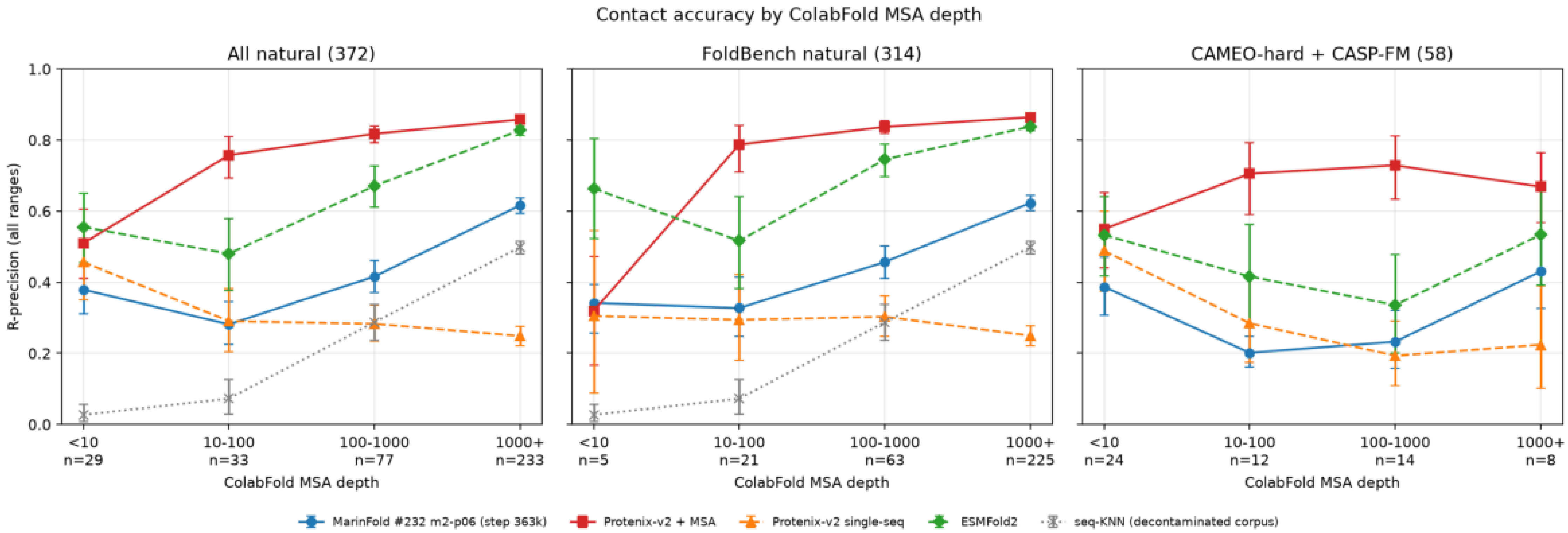
- The **gap to Protenix-v2 + MSA is narrowest where MSAs are thinnest** — −0.131 [−0.232, −0.030] at depth <10 against −0.242 at ≥1000. It never closes, and the narrowing is mostly Protenix falling rather than MarinFold rising. - On **AUC**, MarinFold is the **best predictor measured in the <10 bin** — 0.881, +0.054 [+0.002, +0.105] over Protenix-v2 + MSA and +0.080 over ESMFold2. The shallow-tier R-precision gap is a failure to concentrate contacts in the top-L cut, not a failure to rank them.

Caveats

- The FoldBench half has only 5 proteins under depth 10; the pooled row is the one to read, and it exists because the 58 CAMEO-hard / CASP-FM targets (24 of them under depth 10) were brought in. - Baseline AUC is published for `eval-val` only, so the paired AUC comparison covers 155 of the 372 natural proteins. - Median length is 148 residues in the <10 bin against 290 at ≥1000, and contact prevalence goes as ~1/L. That bias favours the shallow bins, so the reported decline is conservative.

rprecision_by_depth_tier.png

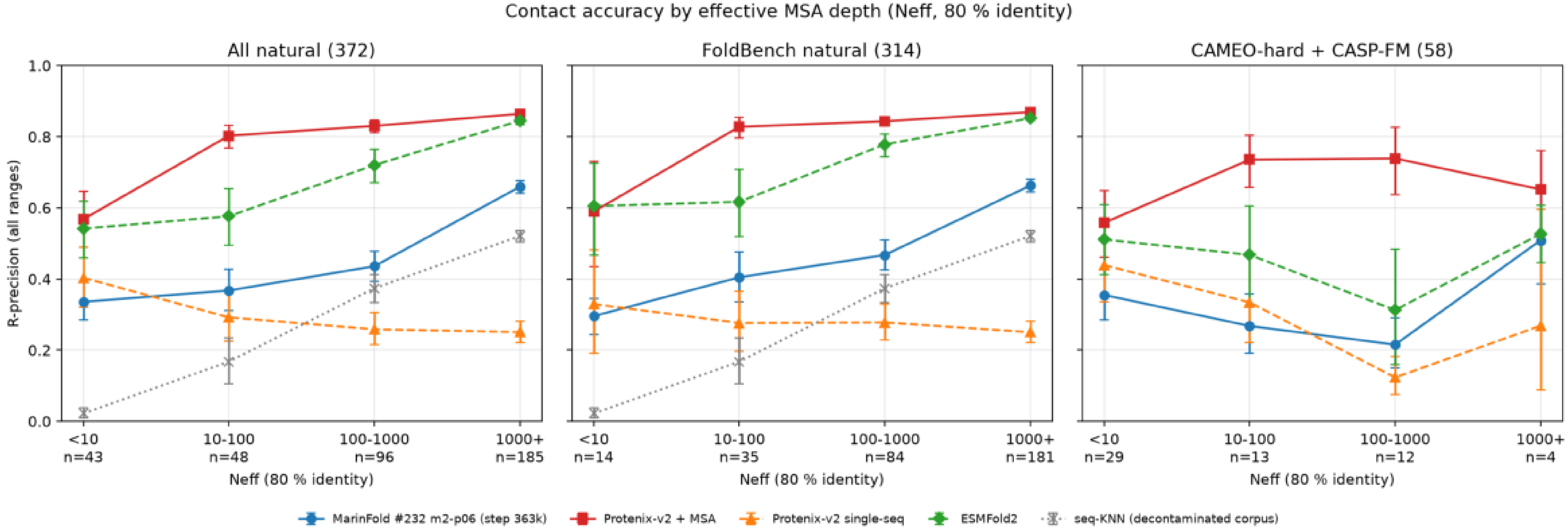
Mean all-range R-precision per MSA-depth tier, with 95 % bootstrap intervals over proteins and bin sizes on the x axis.



$ plot_depth.py

rprecision_by_neff_tier.png

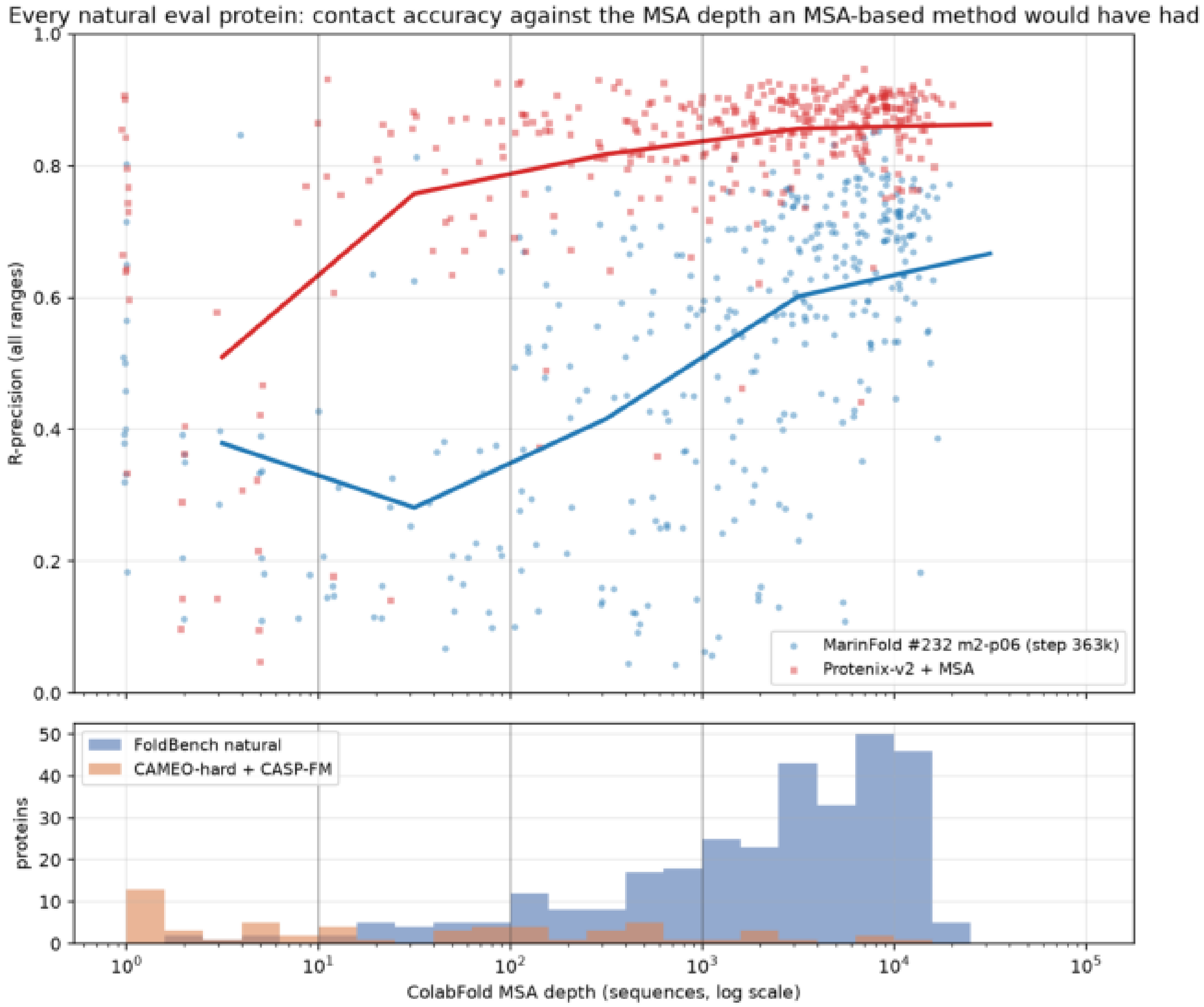
The same cut on redundancy-weighted depth, which fills the shallow bins raw sequence count leaves nearly empty.



$ plot_depth.py

rprecision_vs_depth_scatter.png

Every natural eval protein, accuracy against depth, with per-decade means; the histogram shows where each subset sits.



$ plot_depth.py